EXPERIMENT 4

Design of an Experiment about Fundamental Operations with Voltages

GROUP NO: 11.3

Objectives: In this experiment, design of an experiment about analysis and modification of a circuit, capable of performing the four fundamental operations with voltages, will be investigated.

Materials:

Breadboard

DMM (Digital multi-meter)

7 AC Voltage Source

Oscilloscope

8 Switches

22 Resistor(s)

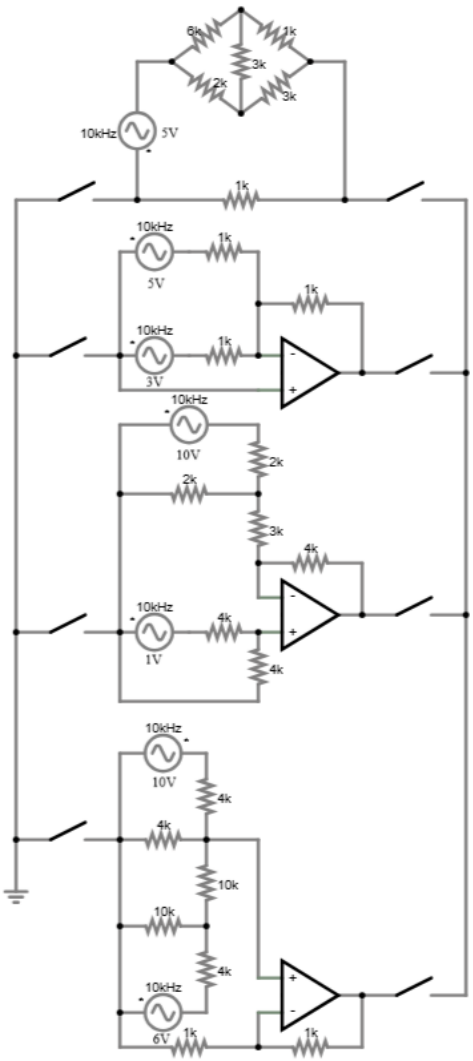
3 LM741s

Division

Addition

Subtraction

Multiplication



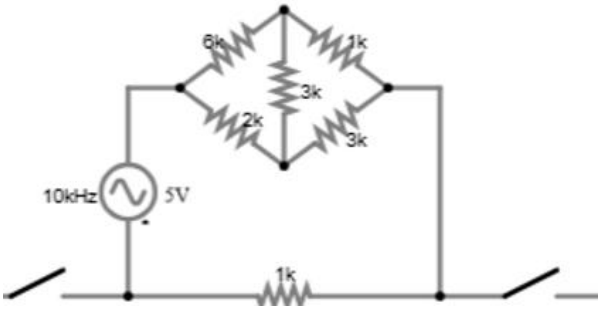
Procedure:

1. In the division part perform a Delta-to-Y transformation for the Wheatstone bridge at the top of the circuit.

   Calculate the equivalent resistances for the Y configuration.

   Simplify the circuit using the calculated values.

   Simulate the original and transformed circuits to verify their equivalence.



2. Determine the output voltage by setting up the summation section in the simulation.

   Identify the voltage at the inverting and non-inverting terminals of the operational amplifier.

   Compare simulated values with theoretical calculations.

3. Apply Thevenin Theorem to the subtraction part and find the simplest circuit design for it.

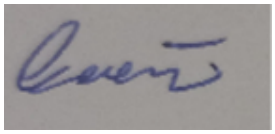
4. Draw the voltage graph of the multiplication part. Simplify the circuit and simulate it on a simulation app (LtSpice, Multisim or falstad). (Hint: To simplify, you can use the mesh current method). Then compare your drawing and simulation graph.

Attention: The directions of AC sources are indicated. Be careful during the simulation.

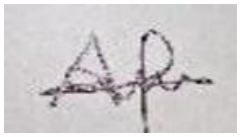
Date: 18.12.2024

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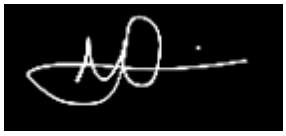
Group Member 1: Kerem Özçakır, 150723025, Developing, modifying and assembling circuits for questions. Report writing. Experiment questions are discussed and created by all group members together.



Group Member 2: Abdulsamet Yaramış, 150722063, Construction of addition and multiplication circuits. Report writing. Experiment questions are discussed and created by all group members together.



Group Member 3: Mert Ali Türkoğlu, 150722056, Construction and modification of the subtraction circuit. Experiment questions are discussed and created by all group members together.



Group Member 4: Göksu Türedi, 150720007, Construction and modification of a division circuit. Experiment questions are discussed and created by all group members together.

